

FLORETION BASE VECTORS, TRIANGULAR TILINGS, AND DIGITWISE S_3 -ACTIONS

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This note studies the signed groups $\mathcal{F}_n$ associated with floretion base vectors of order n . Their positive elements are words in the octal alphabet $\{1, 2, 4, 7\}$, multiplied digit by digit by a local bitwise rule based on XNOR and AND. The case $n = 1$ is the quaternion group Q_8 , while the case $n = 2$ recovers the order-32 floretion group previously studied by Mathar. The construction should not be understood as a new abstract family of finite groups: $\mathcal{F}_n$ is the central product of n copies of Q_8 . Its interest here is instead coordinate-geometric. The floretion encoding gives an explicit digitwise multiplication and a natural map from words in $\{1, 2, 4, 7\}^n$ to a recursive triangular tiling.

The main results are as follows. First, the bitwise rule agrees with quaternionic multiplication in order one and hence, when applied digitwise with signs collected, gives the multiplication of base vectors in every order. Second, the centroid map is equivariant with respect to the digitwise action of S_3 on the digits $\{1, 2, 4\}$ and the corresponding dihedral action on the triangle. Third, the cyclic orbit of every non-unit base vector gives three centroids forming an equilateral triangle. Fourth, the centralizer of a non-unit base vector has cardinality 4^n in the signed group and occupies exactly one half of the positive tiles. Finally, a finite Sierpiński-type support is interpreted as an explicit element of the real algebra generated by the base vectors.

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1. Introduction

Floretons begin from a simple encoding. One takes finite words in the octal digits

$$\{1, 2, 4, 7\},$$

interprets these digits as the quaternionic symbols

$$1 \leftrightarrow i, \quad 2 \leftrightarrow j, \quad 4 \leftrightarrow k, \quad 7 \leftrightarrow e,$$

and multiplies digit by digit, collecting one global sign at the end. In order one this recovers the quaternion group Q_8 , in the tradition going back to Hamilton [12]; for modern introductions to quaternions and finite symmetry groups, see for example [2, 3, 13, 15].

The floretion point of view arose experimentally in the early 2000s, especially in order two. In that setting one observes connections with linear recurrences and integer sequences, some of which are documented in the On-Line Encyclopedia of Integer Sequences and in Munafo's collection of floretion-related sequences [18, 19]. These examples are not needed for the proofs in the present note, but they motivate a more structural formulation.

The order-two case also has an independent group-theoretic history. Mathar studied the order-32 floretion group, identifying it with a group in the GAP library and describing conjugacy classes, normal subgroups, representations, and relations with Q_8 [17]. The present note recovers that case as $n = 2$, but places it in a uniform family of arbitrary order n , using a digitwise encoding and a geometric map to a recursive triangular tiling.

It is important to delimit the novelty correctly. The signed group of order n is not a new abstract finite group. It is the central product of n copies of Q_8 , or, equivalently, the extraspecial 2-group obtained from that central product. Thus the abstract group belongs to standard finite group theory [9, 11]. The contribution of the note is the coordinate model: the bitwise rule makes multiplication explicit, while the centroid map connects the words in $\{1, 2, 4, 7\}^n$ with a recursive triangular tiling. In these coordinates, digitwise group actions and centralizers acquire a simple geometric form.

The results proved below can be summarized as follows.

1. The local bitwise multiplication rule based on XNOR and AND coincides with quaternionic multiplication in order one. Since multiplication in order n is digitwise, this gives the rule in every order.
2. A centroid map

$$P : \Delta_n \longrightarrow \mathbb{R}^2$$

is defined and shown to be equivariant with respect to the digitwise action of S_3 on $\{1, 2, 4\}$ and the dihedral action on the triangle.

3. If $b \neq 77 \cdots 7$, then the three points

$$P(b), \quad P(\gamma b), \quad P(\gamma^2 b)$$

are the vertices of an equilateral triangle.

4. For every non-unit base vector b , its centralizer in $\mathcal{F}_n$ has cardinality 4^n ; among the positive base vectors, the corresponding tile set has cardinality $4^n/2$.
5. A finite Sierpiński-type support is formalized as an element of the real algebra generated by the base vectors.

2. Base vectors and multiplication

Definition 2.1 (Base vectors). Fix $n \geq 1$. A floretion base vector of order n is a word

$$b = b_1 b_2 \cdots b_n$$

of length n in the alphabet $\{1, 2, 4, 7\}$. The set of all positive base vectors of order n is denoted by

$$\Delta_n = \{1, 2, 4, 7\}^n.$$

The identity element is the word

$$e_n := 77 \cdots 7.$$

Definition 2.2 (Signed group and real algebra). Set

$$\mathcal{F}_n := \{\pm b : b \in \Delta_n\}.$$

This is the signed group of base vectors of order n . We also write

$$\mathcal{A}_n := \left\{ \sum_{b \in \Delta_n} q_b b : q_b \in \mathbb{R} \right\}$$

for the real algebra generated by Δ_n , with product extended bilinearly from the multiplication of base vectors.

Remark 2.3 (Relation with central products). For $n = 1$ the group is Q_8 . More generally, $\mathcal{F}_n$ is the central product of n copies of Q_8 . Explicitly, it is obtained from Q_8^n by quotienting the central subgroup

$$K_n = \{(\varepsilon_1, \dots, \varepsilon_n) : \varepsilon_r \in \{\pm 1\}, \varepsilon_1 \cdots \varepsilon_n = 1\}.$$

Thus

$$\mathcal{F}_n \simeq Q_8^n / K_n, \quad |\mathcal{F}_n| = \frac{8^n}{2^{n-1}} = 2 \cdot 4^n.$$

Equivalently, $\mathcal{F}_n$ is the extraspecial 2-group specified by this central product. This observation places the construction in the usual theory of finite groups; below, however, we use the digitwise description because it is the one that interacts directly with the triangular tiling.

2.1. The local quaternionic rule

We use the two-bit encoding shown in Table 1. In octal notation this is equivalent to the three-bit representation

$$1 = 001, \quad 2 = 010, \quad 4 = 100, \quad 7 = 111,$$

where the XOR of the three bits is 1.

octal digit	three-bit form	two-bit code	quaternionic symbol
1	001	(00)	i
2	010	(01)	j
4	100	(10)	k
7	111	(11)	e

Table 1: Local encoding of the four allowed digits. The two-bit code is the one used in the XNOR/AND rule.

Definition 2.4 (Bitwise rule in order one). Let

$$x = (ab), \quad y = (cd),$$

with $a, b, c, d \in \{0, 1\}$. Define

$$x \cdot y = (-1)^{m+1} (a \odot c) (b \odot d),$$

where $\odot$ denotes bitwise XNOR and

$$m = (b \& c) + ((a \odot b) \& d) + (a \& (c \odot d)).$$

Here $\&$ denotes bitwise AND.

Proposition 2.5 (Local equivalence with quaternions). *The rule of the preceding definition agrees with the quaternionic multiplication table in Table 2.*

$\cdot$	i	j	k	e
i	$-e$	k	$-j$	i
j	$-k$	$-e$	i	j
k	j	$-i$	$-e$	k
e	i	j	k	e

Table 2: Quaternionic multiplication in the convention used in this note.

Proof. The verification is direct on the sixteen possible products. The unsigned part of the product is

$$(a \odot c)(b \odot d),$$

while the sign is determined by the parity of

$$m = (b \& c) + ((a \odot b) \& d) + (a \& (c \odot d)).$$

For example,

$$k \cdot j = (10)(01).$$

The unsigned part is

$$(1 \odot 0, 0 \odot 1) = (0, 0) = i.$$

The three sign contributions are

$$0 \& 1 = 0, \quad (1 \odot 0) \& 1 = 0, \quad 1 \& (0 \odot 1) = 0.$$

Hence $m = 0$, so the sign is $(-1)^{m+1} = -1$, and $k \cdot j = -i$. The remaining cases give exactly Table 2. $\square$

Definition 2.6 (Digitwise multiplication in order n). Let

$$b = b_1 \cdots b_n, \quad c = c_1 \cdots c_n$$

be two base vectors in Δ_n . For each position r , write the local quaternionic product as

$$b_r c_r = \varepsilon_r d_r, \quad \varepsilon_r \in \{\pm 1\}, \quad d_r \in \{1, 2, 4, 7\}.$$

We define

$$bc = \left(\prod_{r=1}^n \varepsilon_r \right) d_1 d_2 \cdots d_n.$$

Theorem 2.7 (Bitwise rule in every order). *Applying the local bitwise rule in each position and collecting the local signs gives exactly the multiplication of base vectors of order n .*

Proof. By Proposition 2.5, the bitwise rule computes each local product $b_r c_r$ in order one, including its sign. The definition of order- n multiplication is precisely to multiply position by position and collect the product of the local signs. Thus correctness in order one implies correctness in every order. $\square$

Example 2.8. For instance,

$$iji \cdot jek = (i \cdot j)(j \cdot e)(i \cdot k) = kj(-j) = -kjj.$$

3. The centroid algorithm

The basic geometric step is a map from base vectors to points of the plane. The point associated with a word will be the centroid of the corresponding triangular tile.

Definition 3.1 (Elementary vectors). Set

$$v(1) = (\cos 330^\circ, \sin 330^\circ),$$

$$v(2) = (\cos 90^\circ, \sin 90^\circ),$$

$$v(4) = (\cos 210^\circ, \sin 210^\circ),$$

and

$$v(7) = (0, 0).$$

Definition 3.2 (Centroid map). Fix an initial scale $d_1 > 0$, and let $b = b_1 \cdots b_n \in \Delta_n$. Define

$$P(b) = \sum_{r=1}^n \sigma_r d_r v(b_r), \quad d_r = \frac{d_1}{2^{r-1}},$$

where $\sigma_r \in \{\pm 1\}$ is the sign accumulated from the digits 7 occurring before position r :

$$\sigma_r = (-1)^{\#\{s < r : b_s = 7\}}.$$

Remark 3.3. The digit 7 produces no displacement of its own, but it reverses the direction of all subsequent displacements. This reversal is essential for compatibility with the recursive triangular tiling.

Definition 3.4 (Orientation of a tile). For $b \in \Delta_n$, set

$$N_{124}(b) := \#\{r : b_r \in \{1, 2, 4\}\}.$$

The tile associated with b is oriented upward if and only if

$$N_{124}(b) \equiv n \pmod{2}.$$

Otherwise it is oriented downward.

Remark 3.5. The map P determines only the centroid. The orientation rule supplies the additional information needed to determine the displayed equilateral triangle.

Proposition 3.6 (Compatibility with the recursive triangular tiling). *Let $T_\emptyset$ be an initial equilateral triangle centered at the origin, and let $R_0 > 0$ be its circumradius. At each step subdivide an equilateral triangle into the four equilateral triangles obtained by joining the midpoints of its sides: three corner triangles, labelled by 1, 2, 4, and the central triangle, labelled by 7. The corner directions are those of the elementary vectors $v(1), v(2), v(4)$.*

Put $d_1 = R_0/2$ and $d_r = d_1/2^{r-1}$. If T_b is the triangle in the recursive tiling determined by the word $b = b_1 \cdots b_n \in \Delta_n$, then the centroid of T_b is $P(b)$. Moreover T_b is oriented upward if and only if

$$N_{124}(b) \equiv n \pmod{2}.$$

Proof. We argue by induction on the length of the word. The empty word corresponds to the initial triangle, whose centroid is the origin. Suppose the centroid of the triangle associated with the prefix $b_1 \cdots b_{r-1}$ is known. At level $r - 1$, the circumradius of the current triangle is $R_0/2^{r-1}$, and the centroid of a corner subtriangle lies at distance

$$\frac{1}{2} \frac{R_0}{2^{r-1}} = \frac{R_0}{2^r} = d_r$$

from the centroid of the current triangle, in the direction of the corresponding vertex.

At step r , the directions have been reversed once for each earlier occurrence of the central digit 7. Hence the direction sign is

$$\sigma_r = (-1)^{\#\{s < r: b_s = 7\}}.$$

If $b_r \in \{1, 2, 4\}$, the corresponding corner choice adds the displacement $\sigma_r d_r v(b_r)$. If $b_r = 7$, the central subtriangle is chosen: the centroid is unchanged, but the orientation is reversed for subsequent choices. Summing the contributions over $r = 1, \dots, n$ gives the formula for $P(b)$.

Finally, the three corner subtriangles have the same orientation as the current triangle, while the central subtriangle has the opposite orientation. Thus the final orientation is upward if and only if the number of digits 7 is even. Since

$$\#\{r : b_r = 7\} = n - N_{124}(b),$$

this condition is equivalent to $N_{124}(b) \equiv n \pmod{2}$. □

Remark 3.7. The factor σ_r is not needed merely for S_3 -equivariance: the map obtained by putting all $\sigma_r = 1$ would also be S_3 -equivariant. Its role is instead to make the formula compatible with the recursive triangular tiling. The digit 7 selects the central triangle; it produces no displacement of the centroid, but it reverses the orientation for later choices.

Lemma 3.8 (No cancellation of the first nonzero step). *If $b \neq 77 \cdots 7$, then $P(b) \neq 0$.*

Proof. Let m be the first position such that $b_m \neq 7$. The step at position m has length d_m . The sum of the lengths of all subsequent steps is

$$d_{m+1} + \cdots + d_n < d_m.$$

Therefore the first nonzero step cannot be cancelled by the remaining tail of the path, regardless of the later directions. Hence $P(b) \neq 0$. $\square$

4. The digitwise S_3 -action and triangle symmetries

The identification between permutations of the three vertices and the dihedral symmetries of an equilateral triangle is standard; see, for example, [2].

Definition 4.1 (Digitwise action). The group S_3 acts on Δ_n by permuting the digits $\{1, 2, 4\}$ in each position and fixing the digit 7.

Definition 4.2 (Dihedral action). Let D_3 be the symmetry group of the equilateral triangle generated by the three elementary directions $v(1), v(2), v(4)$. We identify D_3 with the group of permutations of these three vectors. If $\pi \in S_3$, let $\rho(\pi) \in D_3$ be the unique linear symmetry of the plane such that

$$\rho(\pi)v(a) = v(\pi(a)) \quad \text{for } a \in \{1, 2, 4\}.$$

Also set $\rho(\pi)v(7) = v(7) = 0$.

Theorem 4.3 (Equivariance of the centroid map). *For every $\pi \in S_3$ and every $b \in \Delta_n$,*

$$P(\pi b) = \rho(\pi)P(b),$$

where πb denotes the digitwise application of π to the word b .

Proof. Write $b = b_1 \cdots b_n$. Since π fixes the digit 7, the positions of the digits 7 in b and in πb are identical. Hence the same coefficients σ_r and the same lengths

d_r occur in the formulas for $P(b)$ and $P(\pi b)$. Using the definition of $\rho(\pi)$, we obtain

$$\begin{aligned} P(\pi b) &= \sum_{r=1}^n \sigma_r d_r v(\pi(b_r)) \\ &= \sum_{r=1}^n \sigma_r d_r \rho(\pi) v(b_r) \\ &= \rho(\pi) \sum_{r=1}^n \sigma_r d_r v(b_r) = \rho(\pi) P(b). \end{aligned}$$

□

Corollary 4.4 (Reflections and transpositions). *The digitwise transposition (24) corresponds to reflection in the axis determined by 1; the transposition (14) corresponds to reflection in the axis determined by 2; and the transposition (12) corresponds to reflection in the axis determined by 4.*

Proof. This is the preceding theorem in the special case where π is a transposition. □

Corollary 4.5 (The cyclic rotation). *Let $\gamma = (124)$, so that*

$$1 \mapsto 2, \quad 2 \mapsto 4, \quad 4 \mapsto 1, \quad 7 \mapsto 7.$$

Then

$$P(\gamma b) = RP(b),$$

where R is the rotation of the plane sending $v(1)$ to $v(2)$, $v(2)$ to $v(4)$, and $v(4)$ to $v(1)$.

5. Cyclic orbits and equilateral triangles

Theorem 5.1 (The cyclic orbit gives an equilateral triangle). *Let $b \in \Delta_n$ be a base vector different from the identity $77 \cdots 7$. Then the three points*

$$P(b), \quad P(\gamma b), \quad P(\gamma^2 b)$$

are the vertices of an equilateral triangle.

Proof. By equivariance, the three points are

$$P(b), \quad RP(b), \quad R^2 P(b),$$

where R is a rotation by 120° or by -120° , depending on the orientation convention. By the no-cancellation lemma, $P(b) \neq 0$. Hence the three points are distinct. Since they are obtained by rotating a nonzero point about the origin through successive angles of 120° , their mutual distances are equal. They therefore form an equilateral triangle. □

Remark 5.2. This result is visually transparent: the digit cycle $1 \rightarrow 2 \rightarrow 4 \rightarrow 1$ is not only a symbolic permutation, but an actual rotation of the triangular tiling.

6. Centralizer tiles

In this section we distinguish carefully between the centralizer in the signed group $\mathcal{F}_n$ and the corresponding set of positive tiles in Δ_n .

Definition 6.1 (Centralizer in the signed group). For $g \in \mathcal{F}_n$, set

$$C_{\mathcal{F}_n}(g) := \{h \in \mathcal{F}_n : hg = gh\}.$$

Lemma 6.2 (Conjugacy classes). *If $b \in \Delta_n$ and $b \neq e_n$, then the conjugacy class of b in $\mathcal{F}_n$ is*

$$\{b, -b\}.$$

Proof. In each coordinate, conjugation by a quaternionic basis element sends the local symbol i, j, k, e either to itself or to its negative. Hence every conjugate of b is necessarily b or $-b$.

It remains to see that both occur. Since $b \neq e_n$, there is a position r with $b_r \in \{1, 2, 4\}$. Choose a base vector $c \in \Delta_n$ whose entries are 7 in all positions except position r , where we choose a quaternionic symbol that anticommutes with b_r . Conjugation by c changes exactly one local sign and therefore produces the global element $-b$. $\square$

Theorem 6.3 (Cardinality of the centralizer). *If $b \in \Delta_n$ and $b \neq e_n$, then*

$$|C_{\mathcal{F}_n}(b)| = 4^n.$$

Proof. The group $\mathcal{F}_n$ has order

$$|\mathcal{F}_n| = 2 \cdot 4^n.$$

By the preceding lemma, the conjugacy orbit of b has cardinality 2. The orbit-stabilizer theorem gives

$$|\mathcal{F}_n| = |\text{Orb}(b)| |\text{Stab}(b)|.$$

The stabilizer of the conjugation action is precisely the centralizer $C_{\mathcal{F}_n}(b)$. Therefore

$$2 \cdot 4^n = 2 |C_{\mathcal{F}_n}(b)|,$$

and hence $|C_{\mathcal{F}_n}(b)| = 4^n$. $\square$

Definition 6.4 (Positive centralizer tiles). For $b \in \Delta_n$, define the positive tile set of the centralizer by

$$C_{\text{tiles}}(b) := \{c \in \Delta_n : cb = bc\}.$$

We also decompose $C_{\text{tiles}}(b)$ according to the sign of the common product:

$$C_+(b) := \{c \in \Delta_n : cb = bc \text{ and } cb \in \Delta_n\},$$

$$C_-(b) := \{c \in \Delta_n : cb = bc \text{ and } cb \in -\Delta_n\}.$$

Thus

$$C_{\text{tiles}}(b) = C_+(b) \sqcup C_-(b).$$

Remark 6.5. The symbols $+$ and $-$ in C_+ and C_- refer to the sign of the common product $cb = bc$, not to the distinction between commutation and anticommutation. The set $C_{\text{tiles}}(b)$ is the centralizer as seen on positive tiles only.

Corollary 6.6 (One half of the positive tiling). *If $b \in \Delta_n$ and $b \neq e_n$, then*

$$|C_{\text{tiles}}(b)| = \frac{4^n}{2}.$$

In particular, the centralizer displayed as a set of positive tiles occupies exactly one half of the order- n tiles.

Proof. Each $c \in C_{\text{tiles}}(b)$ contributes two elements to the signed centralizer, namely c and $-c$. Conversely, every element of $C_{\mathcal{F}_n}(b)$ is of the form c or $-c$ with $c \in C_{\text{tiles}}(b)$. Hence

$$|C_{\mathcal{F}_n}(b)| = 2|C_{\text{tiles}}(b)|.$$

The preceding theorem gives the result. □

Lemma 6.7 (Unsigned part of a product). *For all $b, c \in \Delta_n$, the products bc and cb have the same underlying base vector, possibly with different signs.*

Proof. In order one, the unsigned part of the bitwise rule is

$$(a \odot c)(b \odot d),$$

which is symmetric in the two factors because XNOR is commutative. Applying this in each position gives the claim in every order. □

Corollary 6.8 (Sign test). *Let $b, c \in \Delta_n$. Then c centralizes b if and only if bc and cb have the same sign. If the signs are opposite, then $bc = -cb$.*

Proof. By the preceding lemma, the two products have the same underlying base vector. Thus they can differ only by sign. They are equal exactly when the signs agree, and they are opposite exactly when the signs differ. $\square$

Remark 6.9 (Squares and cancellations). Let

$$X = \sum_{c \in \Delta_n} q_c c \in \mathcal{A}_n.$$

Then

$$X^2 = \sum_{c \in \Delta_n} q_c^2 c^2 + \sum_{\{c,d\} \subset \Delta_n, c \neq d} q_c q_d (cd + dc).$$

By the sign test, for two base vectors c, d , the products cd and dc have the same underlying base vector and may differ only by sign. If $cd = -dc$, the mixed contribution $q_c q_d (cd + dc)$ vanishes; if $cd = dc$, it contributes $2q_c q_d cd$. This gives a natural computational form for the square of a floretion and connects the calculation of X^2 directly to the centralizer tile sets described above.

Definition 6.10 (Sums of centralizer components). For $b \in \Delta_n$, set

$$\Sigma_+(b) := \sum_{c \in C_+(b)} c, \quad \Sigma_-(b) := \sum_{c \in C_-(b)} c, \quad \Sigma_{\text{tiles}}(b) := \sum_{c \in C_{\text{tiles}}(b)} c.$$

These sums are regarded as elements of the real algebra $\mathcal{A}_n$.

Proposition 6.11 (Vanishing of the two signed components). *Let $b \in \Delta_n$ satisfy*

$$b^2 = e_n.$$

Then

$$\Sigma_-(b)\Sigma_+(b) = 0 \quad \text{and} \quad \Sigma_+(b)\Sigma_-(b) = 0.$$

Proof. Write $C^\varepsilon(b)$, with $\varepsilon \in \{+1, -1\}$, for $C_+(b)$ or $C_-(b)$, respectively, and set

$$\Sigma_\varepsilon(b) := \sum_{c \in C^\varepsilon(b)} c.$$

If $c \in C^\varepsilon(b)$, then c centralizes b and

$$cb = bc = \varepsilon d$$

for a unique $d \in \Delta_n$. Since $b^2 = e_n$, we obtain

$$db = \varepsilon cb^2 = \varepsilon c.$$

Moreover d still centralizes b , since $d = \varepsilon cb$, and hence

$$bd = \varepsilon bcb = \varepsilon cbb = \varepsilon c,$$

while $db = \varepsilon c$ as above. Therefore $d \in C^\varepsilon(b)$. Thus right multiplication by b , after forgetting the sign, permutes $C^\varepsilon(b)$. Consequently

$$\Sigma_\varepsilon(b)b = \varepsilon \Sigma_\varepsilon(b).$$

Since all elements in $C^\varepsilon(b)$ centralize b , we also have

$$b\Sigma_\varepsilon(b) = \varepsilon \Sigma_\varepsilon(b).$$

Now put

$$X = \Sigma_-(b)\Sigma_+(b).$$

On one hand,

$$Xb = \Sigma_-(b)(\Sigma_+(b)b) = \Sigma_-(b)\Sigma_+(b) = X.$$

On the other hand,

$$bX = (b\Sigma_-(b))\Sigma_+(b) = -\Sigma_-(b)\Sigma_+(b) = -X.$$

But X centralizes b , because both factors lie in the subalgebra generated by elements that centralize b . Hence $Xb = bX$, and the displayed equalities give $X = -X$. Over $\mathbb{R}$, this implies $X = 0$. The proof of $\Sigma_+(b)\Sigma_-(b) = 0$ is identical, with the order of the two factors reversed. $\square$

Example 6.12. For $b = ii$ in order two,

$$C_-(ii) = \{ie, jk, kj, ei\}, \quad C_+(ii) = \{ii, jj, kk, ee\}.$$

Thus

$$\Sigma_-(ii) = ie + jk + kj + ei, \quad \Sigma_+(ii) = ii + jj + kk + ee.$$

Since $(ii)^2 = e_2$, the preceding proposition gives

$$(ie + jk + kj + ei)(ii + jj + kk + ee) = 0.$$

Figure 1 shows a small decomposition of $C_{\text{tiles}}(b)$ into its two signed components. Figure 2 shows the two components for a higher-order example. These pictures are not needed for the proofs, but they make the geometric structure of the centralizer tiles visible.

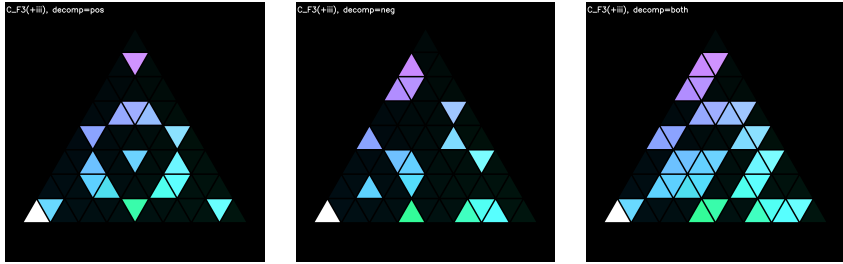
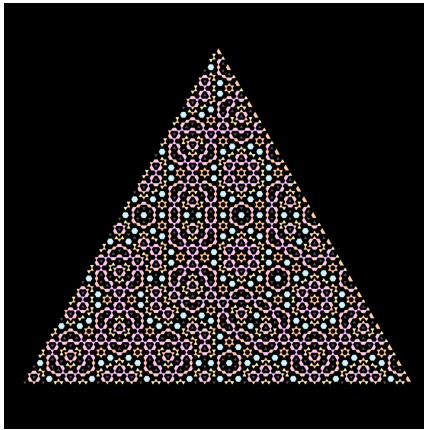
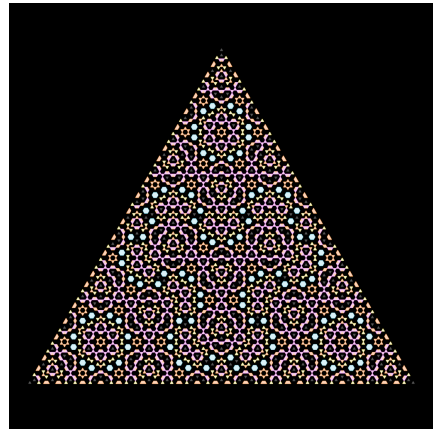


Figure 1: A decomposition of $C_{\text{tiles}}(b)$ into the components $C_+(b)$ and $C_-(b)$ with base vector $b = iii$.



(a) Negative component $C_-(b)$



(b) Positive component $C_+(b)$

Figure 2: The two centralizer components for the base vector $b = ieiiiiii$. The visualizations were generated with the Floretion add-on for Blender.

7. Multiplication by a finite Sierpiński triangle

The author's experimental computations with floretions have led to the informal phrase “multiplying a floretion by a Sierpiński gasket.” The Sierpiński triangle is a classical self-similar set, connected with Sierpiński's original construction and with the modern theory of self-similar sets [10, 14, 16, 20]. In this section we do not use the limiting fractal. Instead, we give the phrase a finite algebraic meaning by interpreting the chosen finite iteration as an explicit element of the algebra $\mathcal{A}_n$.

Definition 7.1 (Discrete Sierpiński support). Define

$$\Delta_n^{\text{Sier}} := \{1, 2, 4\}^n \subseteq \Delta_n.$$

This is the support obtained by eliminating all base vectors containing at least one digit 7.

Definition 7.2 (Sierpiński element). The Sierpiński element of order n is the floretion

$$S_n := \sum_{b \in \Delta_n^{\text{Sier}}} b \in \mathcal{A}_n.$$

More generally, for $a \in \{1, 2, 4, 7\}$, one may define

$$S_n^{(a)} := \sum_{\substack{b \in \Delta_n \\ b \text{ contains no digit } a}} b.$$

The digitwise S_3 -action extends linearly to $\mathcal{A}_n$. The following elementary observation records that S_n is the natural symmetric finite support associated with the three corner digits.

Proposition 7.3 (Symmetry of the Sierpiński element). *For every $\pi \in S_3$,*

$$\pi(S_n) = S_n.$$

More generally, $\pi(S_n^{(a)}) = S_n^{(\pi(a))}$ for $a \in \{1, 2, 4\}$, while $S_n^{(7)}$ is fixed by all of S_3 .

Proof. The set $\{1, 2, 4\}^n$ is preserved by every digitwise permutation of the digits 1, 2, 4. Hence the sum of its elements is fixed. The same argument applies to $S_n^{(a)}$: the condition of avoiding the digit a is carried by π to the condition of avoiding the digit $\pi(a)$. Since every $\pi \in S_3$ fixes 7, the element $S_n^{(7)}$ is invariant. $\square$

Definition 7.4 (Embedding of lower orders). If $m \leq n$ and $u = u_1 \cdots u_m \in \Delta_m$, define the prefix embedding

$$\iota_{m,n}(u) := u_1 \cdots u_m \underbrace{77 \cdots 7}_{n-m \text{ digits}} \in \Delta_n.$$

In particular, a quaternion $q \in \Delta_1$ can be regarded as an order- n base vector by

$$\iota_{1,n}(q) = q77 \cdots 7.$$

Definition 7.5 (Product with a Sierpiński element). Let $x \in \mathcal{A}_n$. To multiply x on the right or on the left by the discrete Sierpiński element of order n means to compute, respectively,

$$xS_n \quad \text{or} \quad S_n x$$

inside the algebra $\mathcal{A}_n$. If x is originally given in order $m < n$, one first chooses an embedding, for instance $\iota_{m,n}$, and then performs the product in $\mathcal{A}_n$.

Example 7.6. The phrase “ i multiplied by the order-five Sierpiński element” can be formalized as

$$\iota_{1,5}(i)S_5 = 17777S_5.$$

This convention is natural but not unique: one could embed i in another coordinate. The choice of embedding is therefore part of the definition of the experiment.

Remark 7.7. This construction does not, by itself, assert that the products obtained in this way form a new finite group. It gives a precise algebraic language for geometric experiments: a finite Sierpiński iteration is not only a picture, but an element of the algebra $\mathcal{A}_n$.

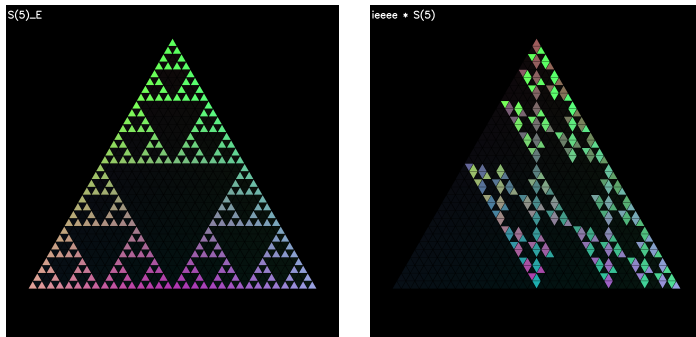


Figure 3: Left: a discrete Sierpiński-type support. Right: a visualization of multiplication by a base vector embedded in higher order.

8. Concluding remarks

The guiding point of the note is that three objects which at first appear separate are tightly connected:

quaternionic multiplication,
recursive triangular tilings,
digitwise permutations of $\{1, 2, 4\}$.

The equivariance theorem identifies the symbolic action with the geometric action, while the centralizer theorem explains why centralizer tiles always occupy exactly one half of the positive tiling. The sign structure also produces a natural simplification in the computation of squares: mixed terms associated with anticommuting pairs cancel.

From a computational point of view, the bilinear product formula can be realized, for each resulting base vector z , as a signed scalar product between the coefficient vector of one factor, reordered according to the bitwise rule, and the coefficient vector of the other factor. For high orders this suggests using precomputed tables of indices and signs, possibly divided into segments, rather than directly multiplying all pairs of base vectors. This observation is not needed for the proofs above, but it helps explain why the bitwise coordinate model is useful in computation.

The definition of the Sierpiński element S_n suggests a further direction: one may study products of geometrically defined elements of $\mathcal{A}_n$ and interpret their images as transformations of the triangular tiling.

A. Historical note on order two

The order-two case was the first to be studied extensively. In the historical notation, the sixteen positive base vectors are written, for example, as

$ie, je, ke, ei, ej, ek, ii, jj, kk, ij, ik, ji, jk, ki, kj, ee,$

together with their signs. Group-theoretically, this signed set has order 32 and is isomorphic to the central product

$$Q_8 \times Q_8 / \{(1, 1), (-1, -1)\}.$$

Mathar analyzed this structure in terms of conjugacy classes, normal subgroups, representations, and its relation with two copies of the quaternion group [17]. In particular, order two in the present construction coincides with the signed order-32 group considered there, while Remark 2.3 gives the uniform form valid for all n .

A different line of investigation, developed in an earlier manuscript of the author [5], treated order-two floretions as elements of a real algebra and studied sequences obtained by inspecting individual base-vector coefficients in the powers X^m . In the most direct form, these coefficient sequences give linear recurrences of order at most four. That setting also contains static and dynamic identities, the so-called Floret’s Cube, and experimental links with Fibonacci, Lucas, and Pell type sequences. Some sequences generated from such experiments have been recorded or discussed in the OEIS; for example, A001109 mentions a “generating floretion” in connection with the so-called “jes” series [19]. These topics are not used in the present note, but they explain some of the historical motivation for the construction.

One future direction is to use the symmetries visible in the triangular tiling as a guide for finding special elements of $\mathcal{A}_n$ and possible identities among sequences extracted from their coefficients.

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REFERENCES

- [1] S. L. Altmann, *Rotations, Quaternions, and Double Groups*, Clarendon Press, Oxford, 1986.
- [2] M. A. Armstrong, *Groups and Symmetry*, Springer, New York, 1988.
- [3] J. H. Conway and D. A. Smith, *On Quaternions and Octonions: Their Geometry, Arithmetic, and Symmetry*, A K Peters, Natick, MA, 2003.
- [4] M. J. Crowe, *A History of Vector Analysis*, University of Notre Dame Press, Notre Dame, 1967.
- [5] C. Dement, *Floretions 2009*, unpublished manuscript, 2010.
- [6] C. K. Dement, *An introduction to Floret’s cube and star via the quaternion product factor space $Q_8 \times Q_8 / \{(1, 1), (-1, -1)\}$* , manuscript, 2004.
- [7] C. Dement, Floretions.com: online calculator and introduction, <https://www.floretions.com>.
- [8] C. Dement, Floretion GitHub repository, <https://github.com/Floretion-Inquisitor/floretions>.
- [9] D. S. Dummit and R. M. Foote, *Abstract Algebra*, 3rd ed., John Wiley & Sons, Hoboken, NJ, 2004.

- [10] K. Falconer, *Fractal Geometry: Mathematical Foundations and Applications*, 3rd ed., John Wiley & Sons, Chichester, 2014.
- [11] D. Gorenstein, *Finite Groups*, Chelsea, New York, 1980.
- [12] W. R. Hamilton, *Elements of Quaternions*, Longmans, Green, London, 1866.
- [13] A. J. Hanson, *Visualizing Quaternions*, Morgan Kaufmann, San Francisco, 2006.
- [14] J. E. Hutchinson, Fractals and self similarity, *Indiana Univ. Math. J.* **30** (1981), no. 5, 713–747.
- [15] J. B. Kuipers, *Quaternions and Rotation Sequences: A Primer with Applications to Orbits, Aerospace, and Virtual Reality*, Princeton University Press, Princeton, NJ, 1999.
- [16] B. B. Mandelbrot, *The Fractal Geometry of Nature*, W. H. Freeman, San Francisco, 1982.
- [17] R. J. Mathar, *Structure of the Floretion Group*, note/preprint, 2010. Available via OEIS at <https://oeis.org/A173343/a173343.pdf>.
- [18] M. Munafo, Sequences related to floretions, <https://www.mrob.com/pub/seq/floretion.html>.
- [19] The OEIS Foundation Inc., The On-Line Encyclopedia of Integer Sequences, Sequence A001109, <https://oeis.org/A001109>.
- [20] W. Sierpiński, Sur une courbe dont tout point est un point de ramification, *C. R. Acad. Sci. Paris* **160** (1915), 302–305.

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